CYS 339: FRAUD AND DETECTION TECHNOLOGIES (2 HRS/WK)

OBJECTIVES:

With today intensive use of integrated systems storing gigantic amount of financial and non-financial data, hardly can any accounting, auditing or investigation work be conducted without data analysis. Valuable evidence may be staring at the accountant and auditor but not picked up if they do not know how to mine and connect the seemingly unrelated data to make sense out of it.

The irreversible trend of Big Data and pressure to remain competitive have compel organization to make use of data analysis and continuous monitoring to understand business trend and consumer behaviours, as well as to detect fraud. Knowing what tool to use to acquire admissible digital evidence and how to search through pile of electronic files and email is a must have skill for today's auditors. This course will expose the learners to various data analysis techniques, tools and templates in the conduct of audit or investigation. The specific objectives for this course include:

- Acquire the overall concept of data analysis in detecting fraud
- Better equipped to identify irregularities in data trend and data profile
- Able to perform qualitative data analysis
- Understand rules governing digital evidence, digital forensic and online investigation
- Understand about procedures in digital forensic and continuous monitoring
- Gain knowledge on common detection techniques of fraud/ wrongdoing

CONTENT

1. **DATA ANALYSIS – OVERVIEW**

- Definition of data analysis
- Three Reason Data Analysis is Must
- Data Analysis in Auditing vs Investigation
- What can Data Analysis do and can't

2. **COMMON ANALYSIS PROCEDURES**

- Financial Ratio, Variance, Percentage, Aging, Average
- Common analysis functions
- Filter / Sort/ Join / Append/ Pivot/ Trend / Regression / Ratio
- Vlookup/ Eye-lookup/ Grouping/ Summarization/ Stratification

3. **TWO TYPE OF DATA ANALYSIS**

- Quantitative – Sorting, Category, Relationship, Profiling
- Qualitative – Contextual, Behavioural

4. PERFORMING DATA ANALYSIS

- Gap analysis
- Duplicate analysis
- Fuzzy duplicate
- Orphan transactions
- Timing analysis
- Limit and Reasonableness
- Trend and Outlier
- Unwanted match
- Aging, Stratification, Sampling

5. PERFORMING DATA PROFILING

- Frequency tests
- Intensity tests
- High risk tests
- Relative Size tests
- Digit Frequency tests (Benford)
- Concentration test (GEL)

6. PERFORMING CONTEXTUAL AND BEHAVIOUR QUALITATIVE ANALYSIS

- Key words
- Content (file, picture), Behaviour Profile
- Online Presence & Social Media
- Digital Forensic and Data Acquisition

7. OLAF APPROACH IN DATA ANALYSIS

- Object Locate Analyse Follow
- Understand “Miscarriage of Justice” and “Impurity of Finding”

8. SUCCESSFUL IMPLEMENTATION OF DATA ANALYSIS

- 3M methodology - Train the Captain, Draw up the Map, Train will Come
- Understand Critical Thinking, Logical Reasoning
- Accounting and Commercial Knowledge
- Data Analysis Tool – Generic vs Specific

9. IMPLEMENTING CONTINUOUS MONITORING AND DIGITAL SURVEILLANCE

- Monitoring software
- Digital forensic acquisition
- Social engineering

10. BEYOND DATA ANALYSIS AND DIGITAL FORENSIC – WHAT NEXT?

- Finding confirmation
- Determine root cause
- Develop Hypothesis
- Move on to interview and interrogation

---------- End of Third Year Courses ----------